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Master Degree in Statistics for Data Science

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Master Thesis

"Forecasting the electricity load daily peaks using support vector machine"

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SUMMARY

This master 's thesis aims to perform Support Vector Machine (SVM) regression for forecasting daily peak electricity load in the New England region. The complexity of predicting peak demand is the interaction between several factors, particularly weather conditions. Using hourly demand data from 2015 to 2023, the study investigates the relationship between temperature and electricity demand across eight load zones, plus the total of New England, and each station is linked to a corresponding weather station.

To address the limitation of uncertainty in future temperature, temperature scenarios were generated using K-Means clustering. These scenarios were then used in to predict the trained SVR models to enhance the robustness. The models were evaluated based on Mean Squared Error (MSE) and Brier Score, with the results compared with three naive forecasting models: Naive24, Naive168, and a Mixed Naive approach.

The findings show that SVR models perform well in most regions, with the scenario-based approach further improving accuracy. Southeastern Massachusetts and Rhode Island exhibited the lowest prediction errors, while regions like Maine proved to be more challenging due to their unique temperature and demand profiles.

Keywords: Electricity load forecasting, Support Vector Regression (SVR), Temperature scenarios, K-Means clustering, Peak Hour prediction, Mean Squared Error (MSE), Brier Score.

DEDICATION

This final master's thesis is the culmination of an incredible year. I have learned so much throughout this journey, and I would like to dedicate this thesis to all those who guided, supported, and encouraged me along the way.

To my tutor, Andrés, your constant guidance, expertise, and knowledge have been essential to the completion of this thesis.

To my friends from the master's program, the Underdogs, thank you for listening, teaching, and supporting me. Our time together will remain in my heart for the rest of my life. Without you, I would not have made it through this year.

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This work is dedicated to all of you who have contributed to this achievement in your unique and meaningful ways. I could not have done it without you.

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1. INTRODUCTION

1.1. Introduction to the Electricity Market

Over the past years, the electricity has been fluctuating due to different factors such as economic activities, social behaviour and weather conditions. Despite some hurdles along the way, the markets have succeeded in providing reliable electricity at the least cost to consumers, which is no simple task. Every second, supply and demand must balance. Thousands of resource and network constraints must be satisfied, and the market must send the right price signals to motivate efficient generation and investment in resources over time. The objective of this section is to explain how the market works, how it has evolved over time, and whether it is suited to handle the large changes that will take place over the future [1].

Electricity markets differs across regions, showing different market settings and path-dependence factors. In this project we will focus on the New England electricity market as we will use its data to exemplify the procedures that we will implement in the following chapters. The electricity market in New England has been changing significantly over the past several decades, evolving from a vertically integrated utility structure to a competitive market designed to ensure reliable and cost-effective electricity supply. This evolution has been driven by the advance of the technology, regulatory reforms and due to economics and technical needs.

In the first years, the New England electricity market was characterized by vertically integrated utilities. This means that they owned and operated all stages of electricity production, distribution, and retail. The utilities operated under rate-of-return regulation, where investments were approved by regulators and costs were passed on to consumers.

The initial step towards market restructuring in New England began in the late 1990s, influenced by broader trends in the U.S. and prompted by the Federal Energy Regulatory Commission (FERC) [2]. The aim was to introduce competition and improve efficiency in the electricity sector. This resulted in the separation of generation, transmission, and distribution functions, and the establishment of wholesale electricity markets.

1.1.1. Creation of ISO New England

ISO New England (ISO-NE) is an independent, not-for-profit organization responsible for managing the electricity grid and wholesale electricity markets in the New England region of the United States. It was created in 1997 to manage the region's bulk power system and wholesale electricity markets. Its primary responsibilities included administering the competitive wholesale electricity markets, ensuring the reliable operation of the electricity grid, and planning for future system needs. ISO-NE operates independently of the electricity generators and utilities, providing a neutral platform to facilitate market transactions. This independence is crucial for maintaining market fairness and transparency [2].

1.1.2. Development of Wholesale Markets

With the creation and establishment of ISO-NE, New England became into a more competitive market structure. The wholesale market was developed to enable real-time trade and pricing of electricity. These are the principal components of the wholesale market:

- **Day-Ahead Market:** In this market is allowed to buy and sell electricity one day before the actual delivery. It helps in securing prices and managing supply and demand expectations.

- **Real-Time Market:** This market operates on a continuous basis, allowing participants to trade electricity in real-time based on actual supply and demand conditions. It is crucial for maintaining grid stability and responding to unforeseen fluctuations.

- **Forward Capacity Market (FCM):** This market is introduced to ensure long-term reliability. The FCM provides incentives for investment in new capacity and ensures that sufficient resources are available to meet future demand.

1.1.3. Nodal Pricing and Market Enhancement

In order to provide more accurate price signals and to address the inefficiencies of the transmission constraints, New England implemented a nodal pricing system. This system is characterized by determining the electricity prices depending on the locations (nodes) within the region. These nodes are not arbitrary points, they correspond to the actual geographic location that reflect the true cost of delivering electricity considering congestion and losses in the transmission network. In addition, to include mechanism for demand response, that allows the consumers to reduce their usage during peaks in response to price signals. For clarity, a map of the New England region with the designated nodes is provided below, which visually represents these locations and their impact on pricing (see Fig.1.1).

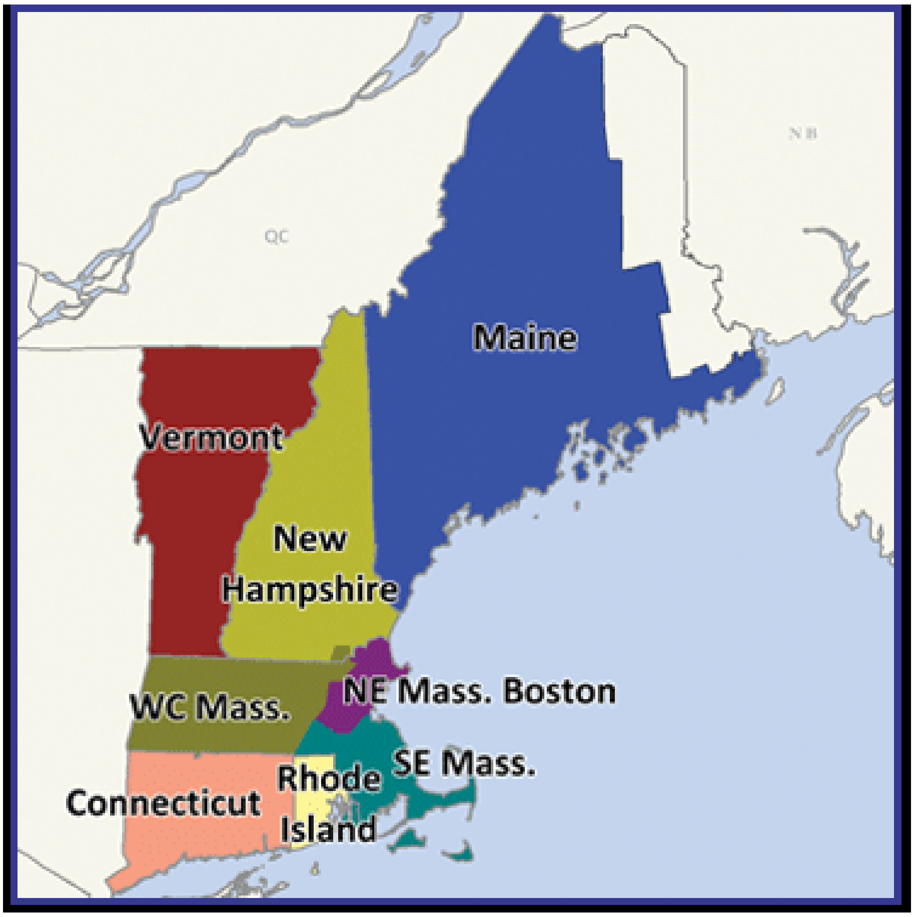

Fig. 1.1. Map of New England regions with designated nodes. Source and Permission granted: https://www.iso-ne.com/.

To better understand the regions within the New England electricity market and their respective designations, Table 1.1 provides a list of the major regions along with their corresponding abbreviations. These abbreviations will be used consistently throughout this study to refer to the respective regions within the electricity market.

Region	Abbreviation
Northeast Massachusetts & Boston	NEMA
Vermont	VT
New Hampshire	NH
Maine	ME
Rhode Island	RI
Southeastern Massachusetts	SEMA
Connecticut	CT
Western/Central Massachusetts	WCMA

TABLE 1.1. NEW ENGLAND REGIONS AND THEIR CORRESPONDING ABBREVIATIONS.

1.2. Peak Hours importance

Peak hours, which are the hours during the day with the highest electricity demand, play a crucial role in the management of electric grids. Understanding and predicting correctly these peak hours is essential for several reasons. During peak hours, the demand for electricity can exceed normal supply capacities. To avoid power shortages, utilities and grid operators must ensure they have sufficient infrastructure and resources available. Accurately forecasting peak hours is useful to optimize the infrastructure investments by reducing the need for expensive and inefficient plants that only operate during peak times. This not only helps in managing operational costs but also ensures that the necessary infrastructure is in place to meet demand without overspending.

Furthermore, peak hour have significant economic implication for consumers, especially for large industrial and commercial users. Many utilities and grid operators implement demand response programs to manage peak load. These programs encourage consumers to reduce their electricity consumption during peak hours by charging higher rates during peaks time. In particular, this programs incentives consumers to shift their electricity usage to off-peak times, which helps to balance the load on the grid. In order to have success on this programs, is crucial to forecast correctly the peak hours [3].

In addition, controlling peak demands is crucial for maintaining the stability and reliability of the grid. High demand during peak hours can exercise pressure over the grid, causing voltage fluctuations and potential equipment failures. Predicting and managing these periods can ensure to grid operators more stable electricity supply and reduce the risk of disruptions. Moreover, minimizing peak demand through better forecasting and demand response, the environmental impact of electricity generation can be reduced, helping to decrease carbon emissions and transition to cleaner energy sources.

1.3. State of the art regarding its prediction

The complexity of predicting peak electricity demand accurately steams from the interaction of various factors, weather conditions, economic activities and unpredictable consumer behaviors. In order to address this challenge, researchers use a range of techniques, and recently more sophisticated, complex and data-driven approaches have been used.

Traditional method for peak load prediction, such as time series analysis using, for example, ARIMA models, Holt-Winter and another regression techniques, have been implemented for their simplicity and effectiveness in capturing long-term trends and seasonal variations in the electricity consumption. Nevertheless, these methods often face difficulties in accurately predicting short-term peaks, particularly when external factors like weather patterns or social events cause sudden demand fluctuations [4].

Various machine learning algorithms have been introduced in order to improve accuracy. For instance, models such as Support Vector Machines (SVM), Random Forests, Gradient boosting (GBM) and Artificial Neural Network (ANN). These models incorporate multivariate predictors, such as weather conditions, calendar information, and economic factors, significantly improving prediction performance over traditional methods like ARIMA, that rely only on univariate historical load data. Advances machine learning methods, enhance accuracy by capturing complex nonlinear relationships in the data. The main idea of these models is to leverage large datasets, combining historical demand with weather and grid data, which help them to capture short-term fluctuations and long-term trends, outperforming traditional time series models [5].

The latest advancements combine traditional statistical methods with machine learning approaches. This technique includes external data sources, such as meteorological information and economic indicators that improves the robustness and accuracy of the predictions. For example, [4] proposed a local prediction approach for forecasting daily peaks loads based on local learning, which reduced the prediction errors by tailoring the model to specific patterns in the data. This technique, used a nonlinear prediction, showing that localized patterns in historical data can be leveraged to enhance forecasting performance.

Shukla and Hong (2024) [6] further advanced this area with their work in the BigDEAL Challenge 2022, which focused on forecasting peak timing. They highlighted the importance of specialized models for peak demand prediction, demonstrating that machine learning models combined with scenario-based approaches—such as temperature forecasting—can improve the accuracy of short-term peak timing forecasts. Their findings emphasized the significance of developing models tailored specifically for peak load and peak timing, as general forecasting models often struggle with these specific challenges

Even though significant progress has been made with the introduction of machine learning model and innovative approaches like local prediction, challenges remain. In specific, in managing the uncertainty introduced by external factors such as extreme weather events, behavioral changes, and the increasing integration of renewable energy sources.

2. METHODOLOGY

2.1. Data overview

The initial phase of this master thesis involves gathering and preparing the datasets. For this study, hourly demand data from 2015 to 2023 was utilized, covering multiple load zones across New England. The demand data was sourced from ISO New England. The datasets were unified into eight separate files, each corresponding to a different load zone. This approach ensures that the electricity demand is analyzed in relation to localized weather conditions for more accurate predictions.

Table 2.1 shows the weather stations, their corresponding load zones, and the weighted contribution of each station to overall demand in the New England region during summer and winter.

Weather Station	Load Zone	NE Summer Weight	NE Winter Weight
Boston	NEMA	20.1%	21.4%
Bridgeport	N/A	7.0%	7.5%
Burlington	VT	4.6%	4.0%
Concord	NH	5.8%	5.5%
Portland	ME	8.5%	8.2%
Providence	RI, SEMA	4.9%	4.8%
Windsor Locks	CT	27.7%	27.7%
Worcester	WCMA	21.4%	20.9%

TABLE 2.1. WEATHER STATIONS.

The first part of this data collection and preparation consisted in ensure that the dataset do not contain any missing values (NAs). The analysis confirmed that all dataset were complete, with no missing values. This step was crucial in ensuring the reliability and accuracy of subsequent analyses. However, certain zero values were observed in the dataset, as seen in Figure 2.1. The drop to zero occurred on March 8, 2015, which corresponds with the start of daylight saving time when clocks are set forward by one hour, resulting in a missing hour in the data. To address this, the demand for the missing hour was imputed using the demand from the previous hour, ensuring continuity and accuracy in the time series for the subsequent analysis.

In order to continue the study, several plots were generated to examine energy consumption trends over time. The graph analysis indicated that, except for Vermont, energy consumption trends were relatively consistent across the locations. Each location showed a stable level of electricity consumption with a noticeable increase at the end of June. The highest demand was observed in July, followed by a decline through September. The peak in September could be explained by different factors such as increased home occupancy post-summer vacations, and seasonal events, including local festivals and agricultural activities.

To simplify the visualization and better highlight the increments observed in June, a consolidated graph representing the aggregate energy demand across all eight zones, referred to as ISONECA, will be presented. Figure 2.1 illustrates the combined demand of the eight zone, providing a clearer view of overall consumption patterns and trends.

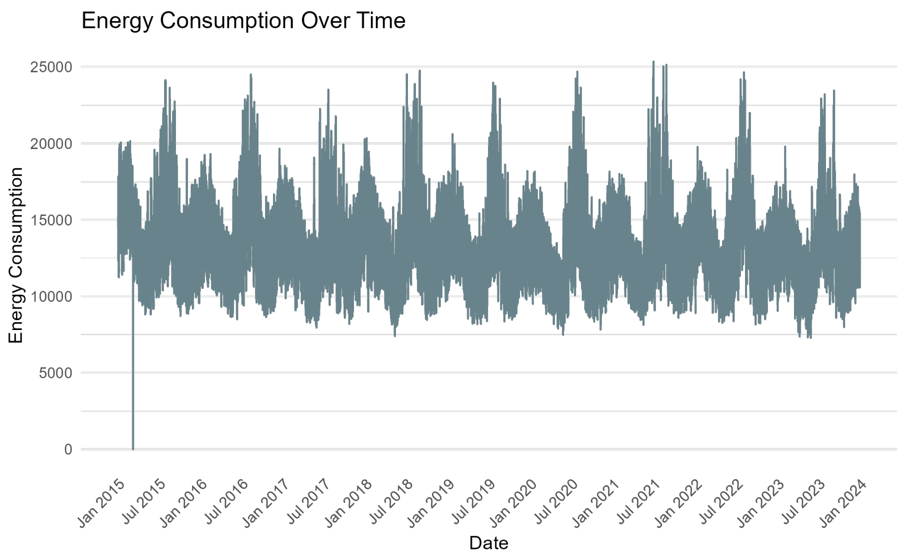

Fig. 2.1. Isoneca Energy consumption over time.

Source: Own elaboration.

Following with the analysis of understand the energy consumption over time, a detailed examination of how energy demand fluctuates across the days of the week and months was conducted for each of the eight load zones. Initially, the data was processed to categorize each date into it corresponding day of the week and month. Then, boxplots were generated to visualize the distribution of daily average demand throughout the week and across different months. The analysis revealed a consistent pattern across all locations: energy consumption tends to be higher on weekdays compared to weekends. This pattern is might be driven by increased commercial and industrial activity during the workweek. Additionally, in most locations, the months with the highest energy consumption were June, July and August, coinciding with the summer season. However, Burlington exhibited a different pattern, where the highest energy demand occurred during the winter months (December, January, and February), likely due to colder temperatures. To simplify the visualization a consolidated of the eight graphs will be presented. Figure 2.2 illustrates the average demand over the days of the week and month.

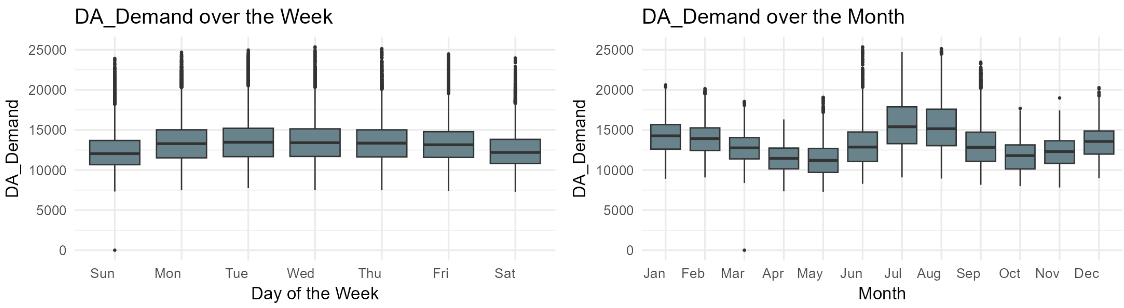

Fig. 2.2. Average Demand over week and month.

Source: Own elaboration.

Initial analyses of overall demand trends revealed consistent patterns, with a notable peak in July and a rise at the end of June. The combined data from all load zones clearly showed these trends.

As it was mentioned in the introduction, the peak hour is the time with the highest demand of the day. The previous analysis of the electricity demand over week and month, was replicated for the most frequent peak hour. The peak hour variable was created using the highest Real-Time Demand (RT_Demand), as DA_Demand may not accurately reflect the actual demand patterns. By using RT_Demand to calculate the peak hour, the variable captures the true demand peaks. The results are presented in the Tables 2.2 and 2.3.

Location	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Maine	18	18	18	18	18	18	18
New Hampshire	18	19	18	19	18	18	18
Vermont	21	20	20	21	21	8	21
Connecticut	18	18	19	18	18	18	18
Rhode Island	18	18	18	18	18	18	18
Southeastern Massachusetts	18	19	19	19	19	19	19
Western/Central Massachusetts	18	18	18	18	18	18	18
Northeast Massachusetts	19	18	18	18	18	18	19

TABLE 2.2. PEAK HOURS BY DAY OF THE WEEK FOR EACH REGION.

Location	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Maine	18	18	20	20	21	19	18	18	20	19	18	18
New Hampshire	18	19	20	20	20	18	18	18	18	19	18	18
Vermont	18	19	20	21	21	21	21	21	20	19	18	18
Connecticut	18	19	20	21	21	18	18	18	20	19	18	18
Rhode Island	18	19	20	21	21	17	17	18	20	19	18	18
Southeastern Massachusetts	18	19	20	20	21	19	18	18	20	19	18	18
Western/Central Massachusetts	18	19	20	20	21	18	18	18	20	19	18	18
Northeast Massachusetts	18	18	19	20	21	18	17	18	18	19	18	18

TABLE 2.3. PEAK HOURS BY MONTH FOR EACH LOAD ZONE.

In these tables, the values represent the most frequent peak hours observed for each day of the week and each month, respectively. The peak hours presented in the table were calculated by determining the mode, which is the most frequently occurring value, of the peak hours for each day and each month. This approach provides insights into the common peak hours and highlights how these peak hours can shift over the course of the week and month.

The analysis demonstrates the significance of weather factors in understanding electricity demand. By correlating demand with localized weather conditions, it becomes evident that weather significantly affects the form of consumption trends, providing valuable insights for more accurate forecasting and demand management.

2.2. Local Approach for SVM Training

Following the study by El-Attar *et al.* [4], a local approach for Support Vector Machine (SVM) training was used to predict the Peak Hour for electricity demand. Historical demand and meteorological data were used to train and validate the model in the different regions. The `eps-regression` SVM was implemented because it captures non-linear relationships and handles high-dimensional data effectively.

2.2.1. Support Vector Regression

Support Vector Regression (SVR), as discussed in the comprehensive tutorial by Smola and Schölkopf [7], is a supervised learning algorithm adapted from SV Machine for regression problems, where the objective is to predict a continuous outcome. This technique has been utilized in several works that required predictive modeling of complex and non-linear relationships, such as forecasting electricity load peaks.

SVR works by mapping input data into a high-dimensional feature space through a kernel function, allowing it to handle the non-linearity efficiently. It utilized an input-output data $\{x_i, y_i\}_{i=1}^{N}$, where $x_i \in \mathbb{R}^n$ represents the input vector and $y_i \in \mathbb{R}$ represents the target value, the objective of SVR is to determine a function $f(x)$ that deviates by no more than ϵ from the actual target values y_i across all training data. This functions is formulated as following:

$$f(x) = w \cdot \phi(x) + b. \tag{2.1}$$

Here, $\phi(x)$ represents a mapping from the input x_i into a higher-dimensional feature space, while w is the weight vector that needs to be estimated, b is the bias term. This optimization problems aims to minimize the complexity of the model (the size of w) while ensuring that the prediction error for each training stays within the margin of ϵ.

To optimize $f(x)$, is necessary to solve the following optimization problem:

$$\min_{w,b,\xi_i,\xi_i^*} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}(\xi_i + \xi_i^*) \tag{2.2}$$

subject to the constraints:

$$\begin{aligned} y_i - w \cdot \phi(x_i) - b &\leq \epsilon + \xi_i^* \\ w \cdot \phi(x_i) + b - y_i &\leq \epsilon + \xi_i \\ \xi_i, \xi_i^* &\geq 0. \end{aligned} \tag{2.3}$$

In this problem formulation, the terms ξ_i and ξ_i^* represents the slack variables, which measure the extent to which prediction fall outside of ϵ-insensitive margin. The constant C controls the balance between the flatness of the function and the amount of tolerated prediction error.

Rather than solving for w directly, it is computationally efficient to work with the dual problem, which is defined as:

$$\min_{\alpha_i, \alpha_i^*} \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} (\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*) K(x_i, x_j) + \epsilon \sum_{i=1}^{N} (\alpha_i + \alpha_i^*) - \sum_{i=1}^{N} y_i (\alpha_i - \alpha_i^*) \tag{2.4}$$

subject to:

$$\sum_{i=1}^{N} (\alpha_i - \alpha_i^*) = 0, \quad 0 \leq \alpha_i, \alpha_i^* \leq C. \tag{2.5}$$

In this dual problem formulation, α_i and α_i^* are introduced as Lagrange multipliers, which are optimized to find the support vectors. The kernel function $K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$ is introduced to handle the non-linear transformation of the input space. This kernels plays a crucial role in SVR, allowing the model to capture non-linear relationships between input variables without computing the feature space mapping ($\phi(x)$) explicitly. In this master thesis the kernel utilized was Gaussian (RBF) kernel:

$$K(x_i, x_j) = \exp\left(-\frac{||x_i - x_j||^2}{2\sigma^2}\right), \tag{2.6}$$

where σ controls the spread of the Gaussian kernel, helping to manage the non-linearity and complexity of the feature space.

By solving the dual problem, the final regression function is:

$$f(x) = \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) K(x_i, x) + b. \tag{2.7}$$

This approach allows for capturing complex, non-linear dependencies between variables such as weather conditions and electricity demand. As El-Attar *et al.* [4] have demonstrated, this localized model provides enhanced accuracy in predicting peak electricity load by focusing on specific weather stations and incorporating temporal and meteorological features.

2.2.2. Model training and prediction

The first step to model train and predict was to prepare the data. Following the study made by Hong *et al.* [8], the first step in this section involved adding several dummy variables to the dataset to build a more comprehensive model. Details of the created dummy variables are outlined below:

- **Day-of-Week Dummies:** Six dummies were created for each day of the week (Monday to Saturday), assuming Sunday as the reference day. This helps to capture daily patterns in electricity consumption that may vary by day, as it was observed before.

- **Month Dummies:** Eleven dummies were created for each month (January to November), using December as the reference month. This is useful to take into account the seasonal effects, which are crucial in understanding demand patterns.

- **Holiday and Pandemic Dummies:** A binary indicator for public holidays was added to isolate any deviations in electricity demand on holidays from regular days. In addition, a "pandemic" dummy variables was included to capture the period during the COVID-19 (March 2020 - December 2020), which significantly impacted economic activity and, consequently, energy consumption.

After creating the dummy variables, in order to accurately predict the daily peak hour, a set of relevant predictors were selected based on historical demand, temperature, and temporal indicators. These features included:

- DA_Demand_hour1 to DA_Demand_hour24: Demand values for each of the previous 24 hours.

- Dry_Bulb_hour1 to Dry_Bulb_hour24: Temperature values (dry bulb) for each of the previous 24 hours.

- $Peak\_Hour\_DayBefore$: The peak hour demand of the previous day.

- $Peak\_Hour\_DayOfWeek7$: The peak hour demand from the same day of the previous week.

- Day of the week dummies: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday.
- Monthly dummies: January to November.
- Pandemic: A binary indicator accounting for the impact of the COVID-19 pandemic.
- Holiday: A binary indicator for public holidays.
- Year: The year of observation.

Given these predictors, the relationship between the response variable *Peak_Hour* and the independent variables was modeled using the following formula:

$$\begin{aligned}\text{Peak\_Hour} \sim\ & \text{DA\_Demand\_hour1} + \text{DA\_Demand\_hour2} + \cdots + \text{DA\_Demand\_hour24} \\ & + \text{Dry\_Bulb\_hour1} + \text{Dry\_Bulb\_hour2} + \cdots + \text{Dry\_Bulb\_hour24} \\ & + \text{Peak\_Hour\_DayBefore} + \text{Peak\_Hour\_DayOfWeek7} \\ & + \text{Monday} + \text{Tuesday} + \cdots + \text{Saturday} \\ & + \text{January} + \text{February} + \cdots + \text{November} \\ & + \text{Pandemic} + \text{Holiday} + \text{Year}.\end{aligned}$$

The above formula captures all 24 demand hours and temperature data, as well as the dummies for days, months, holidays, and pandemic-related effects. The main idea of selecting this predictors were to capture both short-term and long-term that influences on peak electricity demand.

The SVR model was trained using the previous selected features, and was performed for each of the load zone datasets. Each model was trained on data from 2015 to 2021, with the testing period from 2022 to 2023. As it was mentioned before, the kernel used in these models was Radial Basis Function.

Finally, the predictions were calculate for the test period, and the results were evaluated in terms of their accuracy in forecasting the daily peak hour, following the approach outlined by El-Attar *et al.* [4].

For this model, the error measures utilized were the MSE. The **Mean Squared Error (MSE)** is the most common metric used to evaluate the accuracy of a model's predictions.

It measures the average squared difference between the actual values (y_i) and the predicted values ($\hat{y}_i$) [9]. The MSE is sensitive to large errors because it squares the differences, making it more suitable for models where large deviations are undesirable.

Mathematically, the MSE is defined as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(A_i - F_i)^2,$$

where:

- n is the number of observations to be predicted,
- A_i represents the actual observed values,
- F_i represents the forecasted values by the model,

Even though the most common error utilized was the Mean Squared Error (MSE), an adjusted version of this MSE was implemented, this error allows a margin of ±1 hour when evaluating the peak hour predictions. This means that if the predicted peak hour was ±1 of the actual hour, it was considered correctly predicted as well. Since slight deviations of one hour in the forecast can be acceptable, this approach avoids penalizing the model excessively for minor shifts in timing. Allowing an ±1 hour error margin helps capture practically useful predictions, even if not perfectly accurate, as predicting the surrounding peak hours provides valuable insights for grid management.

In addition, several benchmarks were employed for comparison to evaluate the performance of these predictions. The benchmark models employed were:

- Naive24: This model predicts the Peak Hour based on the Peak Hour of the previous day (Peak_Hour_DayBefore).
- Naive168: This model predicts the Peak Hour using the Peak Hour from the same day of the previous week (Peak_Hour_DayOfWeek7).
- Mixed Naive: A hybrid model, which predicts the Peak Hour using the Peak_Hour_DayBefore for Tuesday to Friday and for Sunday, but switches to Peak_Hour_DayOfWeek7 on Mondays and Saturdays.

The MSE with ±1 hour error margin of these benchmark models was computed to compare their performance against the predictions obtained by the SVR model, and the result will be in the Results section.

2.3. Scenarios Generation for Exogenous Variables

In the earlier sections of this final master thesis, real temperature data reported from the weather stations was used to train and predict the models. However, an important limitation that can affect the accuracy and reliability of forecasts is that real temperatures are unknown when making predictions for future periods.

A possible solution to this limitation was to generate temperature scenarios based on historical data. By creating scenarios using clustering techniques from past temperature profiles, it is possible to simulate future conditions more effectively. This approach is helpful to explore how different temperature patterns might impact the model's predictions.

2.3.1. K-Means Clustering for Temperature Profiles

K-Means clustering is used to identify distinct temperature profiles from historical data by partitioning it into k clusters. The goal is to minimize the within-cluster sum of squares (WCSS), which measures the variance within each cluster. This is achieved by iteratively assigning data points to the nearest cluster centroid and then updating the centroids based on these assignments [10].

The objective function for K-Means clustering is defined as:

$$\text{minimize} \quad SCDG = \sum_{g=1}^{G} \sum_{j=1}^{p} \sum_{i=1}^{n_g} (x_{ijg} - \bar{x}_{jg})^2, \tag{2.8}$$

where:

- $SCDG$ denotes the sum of squares within groups,
- x_{ijg} is the value of the j-th variable for the i-th element in group g,
- $\bar{x}_{jg}$ is the mean of the j-th variable in group g,

- n_g is the number of elements in group g,
- p is the number of variables.

An equivalent criterion is to minimize the squared Euclidean distance between each point and its cluster center [11]:

$$\text{minimize} \quad \sum_{g=1}^{G} \sum_{i=1}^{n_g} ||x_{ig} - \bar{x}_g||^2, \tag{2.9}$$

where $\bar{x}_g$ is the centroid of group g, and $|| \cdot ||^2$ denotes the squared Euclidean distance. The K-Means clustering algorithm follows these steps:

1. Initialize k cluster centroids randomly.
2. Assign each data point to the nearest centroid based on Euclidean distance.
3. Recalculate centroids as the mean of all data points assigned to each cluster.
4. Repeat steps 2 and 3 iteratively until convergence is achieved, meaning the assignments of data points to clusters no longer change significantly.

This method effectively groups similar temperature profiles, facilitating the generation of scenarios based on historical patterns. The K-Means algorithm helps in understanding different temperature regimes by clustering days with similar temperature characteristics.

To determine the optimal number of clusters, the Elbow method was used. This method consist in calculate the total within-cluster sum of squares (WSS) for different values of k [12]. The WSS measures the compactness of clusters and is defined as:

$$\text{WSS}(k) = \sum_{i=1}^{k} \sum_{x \in S_i} ||x - \mu_i||^2. \tag{2.10}$$

The optimal k is chosen where the curve of explained variation (WSS) shows an 'elbow', indicating diminishing returns with the addition of more clusters.

2.3.2. Temperature Profile Clustering

To accurately analyze and predict the peak hour demand, is important to account the different temperature profile to assessing the temperature scenarios. The K-means clustering algorithm was applied to historical temperature data, identifying the dates with similar temperature profiles. The Elbow method indicated that six clusters provide the optimal partition for each location.

To simplify the visualization, the elbow plot of the ISONECA (Total New England demand) is displayed in Figure 2.3.

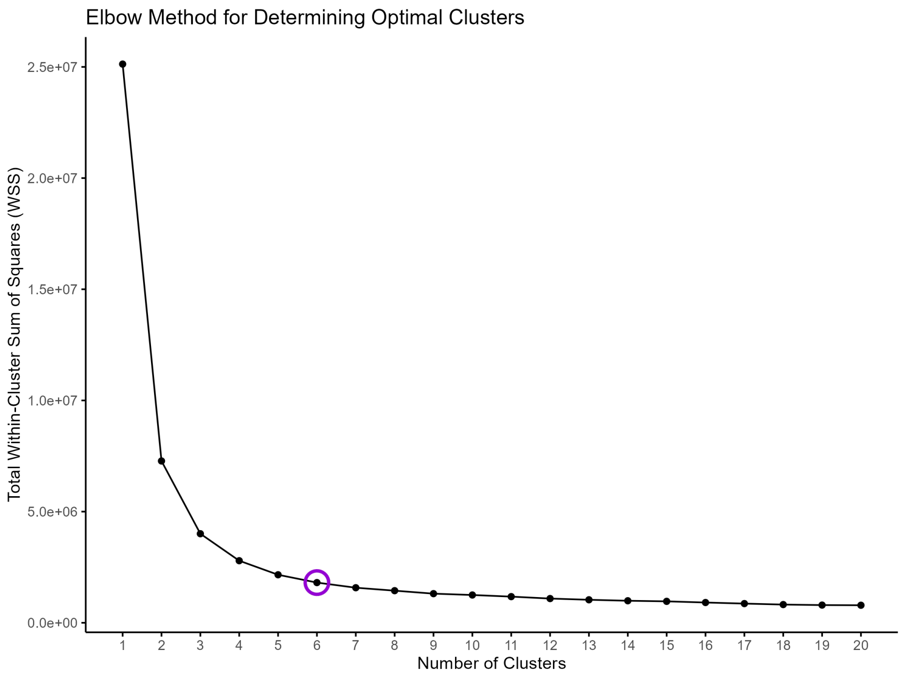

Fig. 2.3. Elbow plot for ISONECA

As is possible to observe, the variation in the shape of the curves is clearly observed across the six clusters, making this the optimal number for implementing the temperature profiles.

After finding the optimal number of cluster, the `temperature_cluster` function was performed for each location. This function groups daily temperature profiles into clusters using K-Means. It operates as follows:

- Converts the date variable to date format.
- Selects and filters hourly temperature data.
- Applies K-Means clustering to assign each day to a cluster based on its temperature profile.
- Adds the cluster labels back to the original dataset.

- Returns the updated dataset and clustering results.

This function helps identify distinct temperature patterns across different days for further analysis.

Figure 2.4 shows the hourly temperature profiles for the six clusters. Each point represents the temperature at a specific hour within a cluster. The plot highlights how daily temperature patterns differ across clusters.

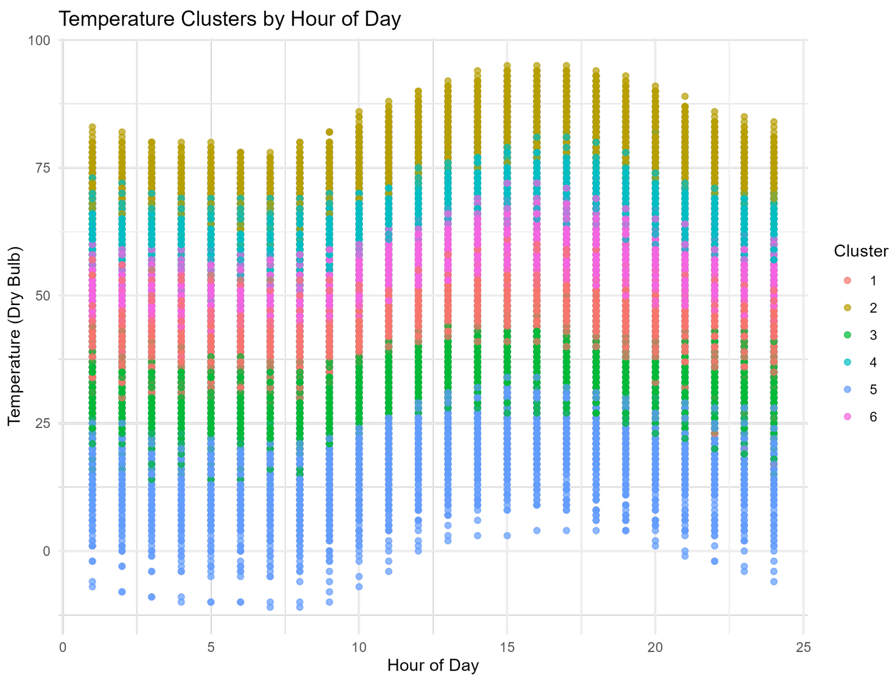

Fig. 2.4. Hourly Temperature Profiles by Cluster for ISONECA

The plot in Figure 2.4 visualizes temperature profiles clustered by hour of the day for the ISONECA region. Each cluster represents a distinct group of temperature patterns throughout the day. The x-axis shows the hours of the day (1 to 24), while the y-axis displays the temperature (Dry_Bulb) in degrees Fahrenheit.

- **Cluster Separation:** The clusters are separated across different temperature ranges. Cluster 1, in light blue, corresponds to the coolest temperature profiles, while Cluster 2, in yellow, includes the highest temperatures.

- **Diurnal Temperature Patterns:** Each cluster exhibits a characteristic pattern over the course of a day. Typically, temperatures rise during the daytime (hours 6 to 18) and fall during nighttime (hours 1 to 6 and 18 to 24). However, the magnitude of these changes differs by cluster.

- **Temperature Range by Cluster:**
  - Cluster 1 represents the lowest temperatures, consistently below 50°F throughout the day.
  - Cluster 2 represents the highest temperature profiles, ranging from 70°F to 100°F.
  - The remaining clusters (3 to 6) fall between these two extremes, with varying degrees of temperature fluctuation.

Even though some variability in temperatures within each cluster at specific hours can be observed, indicating that although clustering captures broad similarities, individual days within the same cluster can still exhibit some variations.

This figure is a representation of the same plot for each location, so is possible to extrapolate the insight to the other locations.

2.3.3. Generating Temperature Scenarios

Once the clusters were identified, they were used to generate temperature scenarios for future predictions. By sampling historical days from similar clusters, several temperature profiles were created for future periods. These scenarios helps to evaluate the impact of different temperature patterns on peak hour electricity demand forecasts, improving the model's overall performance.

The process of generate the temperature scenarios involves:

1. Selecting days from the historical dataset that belong to the same cluster as the target day.
2. Randomly sampling five neighbor days from these similar days.
3. Creating temperature profiles for each sampled day and applying shifts by ±1 hour to simulate variations.

Below is the pseudo-code for the function that generates these scenarios:

Algorithm 1 Generate Scenarios Function

function GENERATE_SCENARIOS(data, current_date, temp_prefix, date_var, cluster_var, n_neighbors, hour_range)

Ensure *cluster_var* exists in data, otherwise **terminate**.

Convert *date_var* to date format.

Identify *target_cluster* from *current_date*.

if *target_cluster* is empty **then**

Terminate.

end if

Sample *n_neighbors* from *cluster_dates*.

Generate scenarios by filtering data for each date and selecting *temp_prefix* columns.

Shift hours using modulo operation for all shifts within *hour_range*.

Create dataframes for each shifted scenario and update with new hour values.

Return combined scenario dataframes.

end function

The temperature scenarios for the years 2022-2023 were generated using the shifted-date method. The generated scenarios were then used to forecast the peak hour electricity demand for the target period.

The model performance was evaluated using two metrics: **Mean Squared Error (MSE)** and the **Brier Score**. The MSE was used to measure the accuracy of point forecasts, while the Brier Score was applied to assess the probabilistic forecasts of peak timing. The MSE error was calculated using the most frequent peak hour predicted in the fifteen different scenarios generated. The Brier Score is a proper scoring rule for evaluating the skill of probabilistic forecasts, particularly for binary outcomes. It measures the squared distance between the forecasted probability (f_t) and the actual outcome (o_t) for each observation over N observations. The Brier Score formula is given by:

$$\text{Brier Score} = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2.$$

This metric is described in the work by Shukla and Hong [6]. In this context, the forecasted probability for each hour representing the daily peak was derived from the frequency of that hour appearing as the peak across different temperature scenarios.

The results will be presented in the following section.

3. RESULTS

This final section represents the results obtained using **Support Vector Regression** for predicting the peak hour electricity demand. The section would be separated in two subsections, the first one corresponds to the prediction results of the SVR using the temperature (Dry_Bulb) corresponding to each load zone. The second one will consist in the prediction results of the models with the fifteen different temperature scenarios generated.

3.1. Standard SVR results

The Table 3.1 summarizes the error metrics obtained from the Support Vector Regression (SVR) model using the temperature (Dry_Bulb) variable corresponding to each weather station.

The column **MSE** represents the model's performance after applying the ±1 hour adjustment, while the **MSE** in the Table 3.2 reflects the regular MSE (without the margin). Additionally, each table includes comparisons with three naive benchmarks: **Naive24_MSE**, **Naive168_MSE**, and **NaiveMixto_MSE**. These benchmarks correspond to simple models: **Naive24_MSE** uses the peak hour from the previous day, **Naive168_MSE** uses the peak hour from the same day of the previous week, and **NaiveMixto_MSE** combines these two approaches. These benchmarks provide useful references for assessing the SVR model's effectiveness.

Location	MSE	Naive24_MSE	Naive168_MSE	NaiveMixto_MSE
Total New England	0.27	0.13	0.15	0.13
Northeast Massachusetts and Boston	0.20	0.27	0.30	0.27
Southeastern Massachusetts	0.05	0.10	0.12	0.11
Connecticut	0.09	0.16	0.20	0.16
New Hampshire	0.18	0.23	0.22	0.22
Vermont	0.12	0.20	0.19	0.20
Rhode Island	0.13	0.19	0.22	0.20
Maine	0.31	0.23	0.24	0.23
Western/Central Massachusetts	0.10	0.18	0.20	0.18

TABLE 3.1. MODELS PERFORMANCE USING MSE WITH ±1 HOUR ADJUSTMENT

Location	MSE	Naive24_MSE	Naive168_MSE	NaiveMixto_MSE
Total New England	0.52	0.35	0.41	0.35
Northeast Massachusetts and Boston	0.50	0.46	0.50	0.44
Southeastern Massachusetts	0.31	0.34	0.39	0.35
Connecticut	0.39	0.42	0.47	0.42
New Hampshire	0.45	0.42	0.44	0.42
Vermont	0.43	0.41	0.45	0.42
Rhode Island	0.45	0.39	0.47	0.41
Maine	0.62	0.43	0.49	0.46
Western/Central Massachusetts	0.39	0.38	0.45	0.38

TABLE 3.2. MSE WITHOUT THE ±1 HOUR ADJUSTMENT

The performance metrics of the SVR model, presented in Table 3.1, shows, in general, a strong performance across various locations, with `MSE` values ranging from 0.05 to 0.31, indicating that the SVR model performs well. However, the `MSE` in the Table 3.2, which reflects the actual mean squared error without adjustment, ranges from 0.30 to 0.62, demonstrating that calculating the MSE without the tolerance could be quite strict.

The SVR model generally outperforms the naive benchmarks in Table 3.1; for example, in Northeast Massachusetts and Boston, the SVR model's (`MSE`) of 0.20 is lower than the `Naive24_MSE` of 0.27 and the `Naive168_MSE` of 0.30, highlighting its effectiveness in capturing complex patterns beyond these simpler models. In Table 3.2 is different, in general the naive benchamrks have an error lower than the MSE, for example, in Northeast Massachusetts and Boston, the SVR model's (`MSE`) of 0.50 is greater than the `Naive24_MSE` of 0.46 and the `NaiveMixto_MSE` of 0.44. This difference can be because the naive benchmarks are simple repetition of past peak hours, which can be sometimes closely to the actual peak hour, especially when patterns are highly regular. In contrast, the SVR model attempts to learn complex relationships, which may not always perfectly capture the exact peak hour, particularly without the ±1 hour tolerance.

However, in Table 3.1 for the Total New England region, the naive benchmarks outperform the SVR model, with `Naive24_MSE` and `Naive168_MSE` both showing better performance than the SVR's `MSE` of 0.27. Regionally, the SVR model performs well in Southeastern Massachusetts with the lowest `MSE` of 0.05, while the Maine region shows the highest `MSE` at 0.31, suggesting that unique regional patterns might be challenging for the model. The similar situation occurs using the MSE without the tolerance, in Table 3.2 the naive benchmarks performed better than the MSE, for Total New England.

Overall, the SVR model provides a competitive edge over the naive benchmarks in most regions, though the higher errors in certain regions like Maine and the relatively poorer performance in Total New England indicate areas where further model refinement or additional features may be needed.

The bar chart in Figure 3.1 further shows the differences in MSE across seasons and regions. It can be observed that some regions, such as Maine, exhibit significantly higher errors during the winter season, suggesting that seasonality plays a crucial role in the predictive performance of the model. This could be because the variations in temperature and other weather-related factors, such as humidity, wind speed, and precipitation, become more pronounced during winter months, making it harder for the model to accurately predict peak demand.

However, other regions, such as Southeastern Massachusetts, show lower and more consistent errors across both seasons, indicating better model adaptability to varying seasonal conditions. This suggests that the SVR model captures seasonal fluctuations more effectively in some areas, possibly due to more stable weather patterns or better alignment between temperature and electricity demand.

Weather conditions is an important feature in peak hour electricity demand forecasting. Temperature fluctuations, especially during extreme weather conditions, tends to cause changes in electricity consumption, as households and businesses rely on heating or cooling systems. The higher prediction errors in regions like Maine during winter might reflect the model's difficulty in accounting for sudden temperature drops or periods of sustained cold weather, which can significantly increase heating demand.

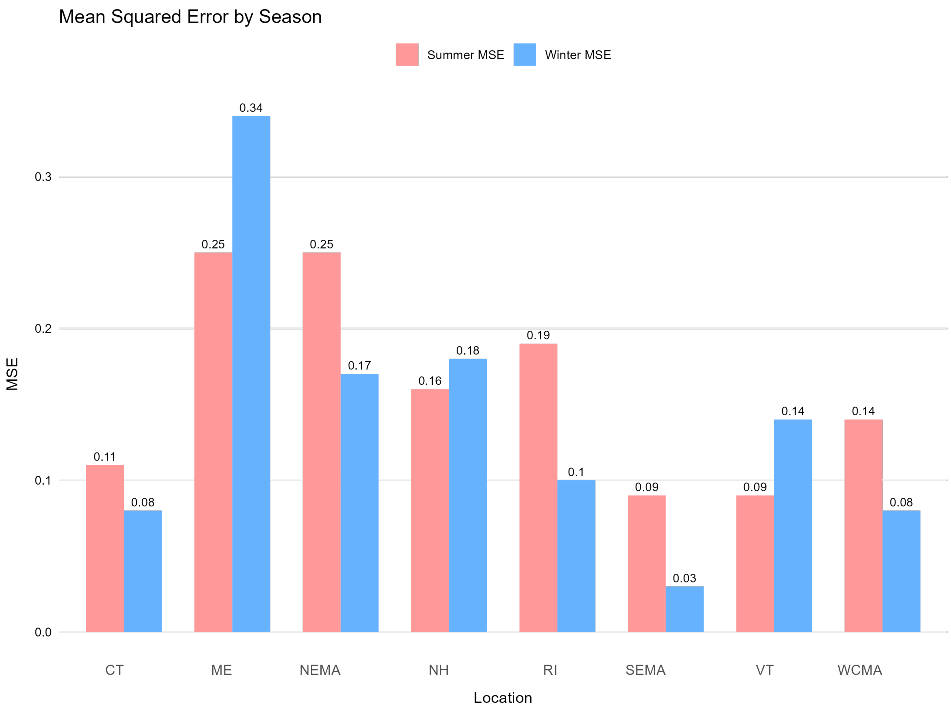

Fig. 3.1. Mean Squared Error with ±1 hour adjustment for the eight load zones in summer and winter

3.2. Temperature scenarios results

As it was mentioned before, the generation of the temperature scenarios is crucial to accurately predict the peak hour electricity demand. Once the temperature scenarios were generated (fifteen for each region), the next step was to evaluate their performance. Table 3.3 presents the results of this evaluation, showing the performance of the SVR model under different scenarios. The table shows the same errors as mentioned before, the `MSE` (allowing for a tolerance of ±1 hour in the predictions), and the `Naive_MSE`. In addition, Table 3.4 shows the Mean Squared Error of the SVR models and the naive benchmarks without the tolerance.

Location	MSE	Naive24_MSE	Naive168_MSE	NaiveMixto_MSE
Total New England	0.08	0.13	0.16	0.13
Northeast Massachusetts and Boston	0.22	0.27	0.30	0.27
Southeastern Massachusetts	0.06	0.10	0.12	0.11
Connecticut	0.10	0.16	0.20	0.164
New Hampshire	0.18	0.23	0.22	0.22
Vermont	0.13	0.20	0.20	0.197
Rhode Island	0.15	0.19	0.22	0.20
Maine	0.33	0.23	0.24	0.23
Western/Central Massachusetts	0.12	0.18	0.20	0.18

TABLE 3.3. SCENARIOS MODELS PERFORMANCE USING MSE WITH ±1 HOUR ADJUSTMENT

Location	MSE	Naive24_MSE	Naive168_MSE	NaiveMixto_MSE
Total New England	0.34	0.35	0.41	0.35
Northeast Massachusetts and Boston	0.53	0.46	0.50	0.44
Southeastern Massachusetts	0.32	0.34	0.39	0.35
Connecticut	0.41	0.42	0.47	0.42
New Hampshire	0.46	0.42	0.44	0.42
Vermont	0.48	0.41	0.45	0.42
Rhode Island	0.47	0.39	0.47	0.41
Maine	0.66	0.43	0.49	0.46
Western/Central Massachusetts	0.41	0.38	0.45	0.38

TABLE 3.4. SCENARIOS MSE WITHOUT THE ±1 HOUR ADJUSTMENT

Analyzing the `MSE` in Table 3.3 across locations, every region performs significantly better than the MSE shown in Table 3.4. The `MSE` ranges from 0.06 to 0.33, indicating that the models perform relatively well. On the other hand, the `MSE` without the tolerance have values from 0.32 to 0.65, almost twice the `MSE` with the tolerance. This proves, one more time, that the `MSE` without the adjustment could be quite strict.

In most regions, the `MSE` in Table 3.3 indicates a strong performance of the scenario-based SVR model. For example, Southeastern Massachusetts is the region with the lowest value (0.06), showing the model's high accuracy in this region when the tolerance is applied. Moreover, Total New England shows a low `MSE` (0.08), suggesting that temperature scenario generation provides well peak hour forecasting across the entire region. However, the Maine region displays the highest `MSE` (0.33), which means that the model has more difficulty in this area, possibly due to unique temperature and demand patterns that are harder to capture accurately through scenario generation. Even though the region with the highest and lowest adjusted errors are noted, it is important to recognize that the `MSE` in Total New England has significantly improved. Generally, the MSE after generated the scenarios do not increase, comparing to the one before generate the scenarios, across the region, showing that generating temperature scenarios may address the limitation of the unknown future temperature without increasing prediction error.

The comparison with the naive benchmarks shows that the models with generated temperature scenarios generally perform better than the naive methods. For instance, in Total New England, the `MSE` of 0.08 outperforms all naive errors, with `Naive24_MSE` at 0.13, `Naive168_MSE` at 0.16, and `NaiveMixto_MSE` at 0.13. This pattern is consistent across most regions, such as Southeastern Massachusetts and Western/Central Massachusetts, where the `MSE` values are lower than the respective naive benchmarks, demonstrating the superiority of the Support Vector Regression models.

However, in regions like Maine, the model's performance is closer to the naive benchmarks, with the `MSE` of 0.33 being comparable to the naive errors (e.g., `Naive168_MSE` of 0.24). This may suggest that, in some regions, additional adjustments may be needed to better capture the complex relationship between temperature and electricity demand.

In contrast, in the Table 3.4, in general, the naive benchmarks outperforms over the SVM models. As it was mentioned before, this can be due to that the naive benchmarks used past peak hours, and this peak hours can be the same as the actual peak hour, or closely to them.

Overall, the SVR model with the generated temperature scenarios shows that this method is more realistic and useful without risking effectiveness.

Figure 3.2 shows the differences in MSE across seasons for each region. It can be observed that some regions, such as Maine, exhibit significantly higher errors during the winter season, suggesting that seasonality plays a crucial role in the predictive performance of the model. Southeastern Massachusetts, however, shows lower and more consistent errors across both seasons, indicating that the model captures better the seasonal conditions. Based on this figure, is possible to deduce that the SVR model captures seasonal conditions more effectively in some areas, this may be because some areas have more stable weather patterns than others.

As it was mentioned before the main insight of this graph is that weather conditions is an important feature in peak hour electricity demand forecasting. Temperature fluctuations, mainly during extreme weather conditions.

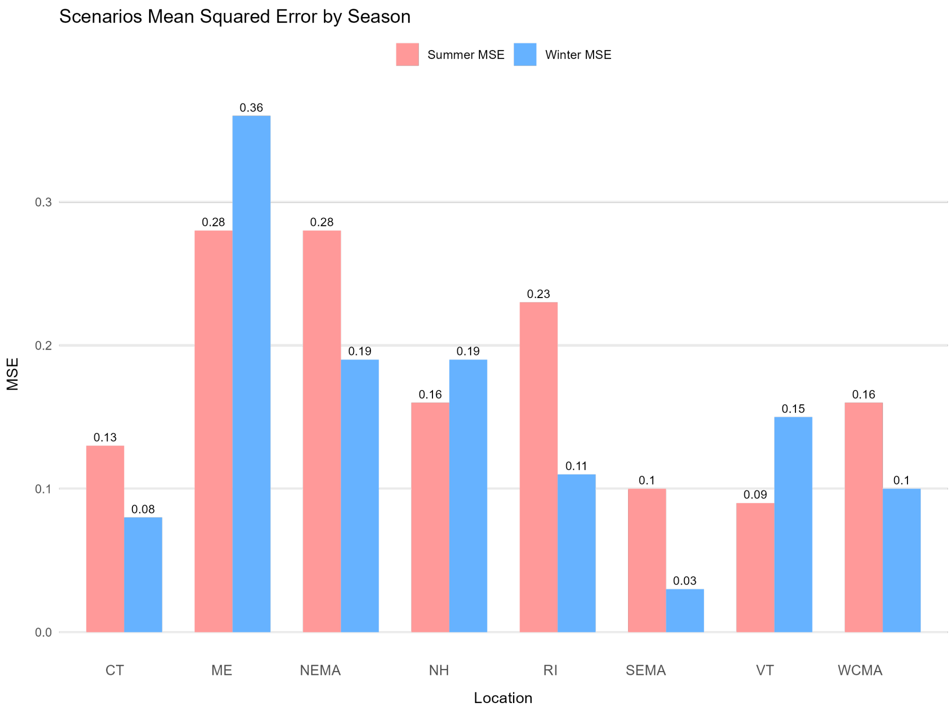

Fig. 3.2. Scenarios Mean Squared Error with ±1 hour adjustment for the eight load zones in summer and winter

Finally, the Brier score was used to assess the accuracy of probabilistic forecasts. As it was mentioned before, it measures the mean squared difference between predicted probabilities and the actual outcomes. A lower Brier score indicates better accuracy of the predictions.

The Brier scores are compared across different regions to highlight the variability in model performance. By examining these results, it is possible to identify regions where the forecasting models performed better and where they may need improvement.

Below, the Brier score results are shown in Figure 3.3.

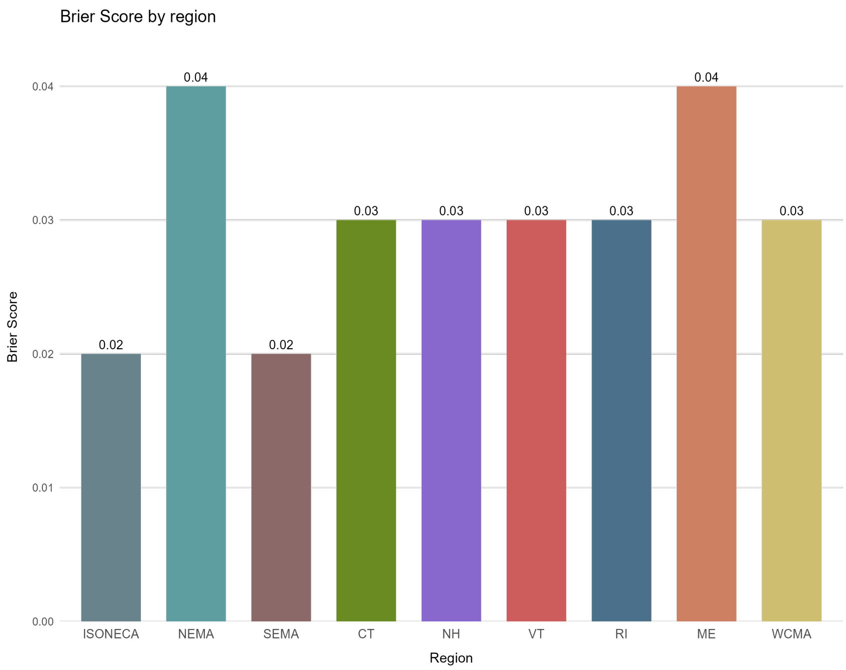

Fig. 3.3. Brier Scores for the eight load zones and the aggregated zone.

Observing the figure, is possible to conclude that in general the Brier score are relatively low. Moreover, Southeastern Massachusetts exhibit the lowest Brier scores, indicating better performance in these regions. On the other hand, the regions with the highest Brier scores are Northeast Massachusetts & Boston, as well as Connecticut. When considering the MSE approach, Maine and Northeast Massachusetts & Boston show the highest errors.

4. CONCLUSION AND FUTURE WORK

This master's thesis presented a broad study on forecasting daily peak electricity load in the New England region using Support Vector Regression (SVR). The main hypothesis was that the interaction between various factors, especially weather conditions, play a crucial role in the electricity demand. The study analyzed hourly demand data from 2015 to 2023 across eight load zones plus the total zone, with each zone linked to a corresponding weather station.

The SVR model predicted effectively peak hour demand, particularly when incorporating temperature data. The introduction of temperature scenarios, generated using K-Means clustering, significantly enhanced the robustness of the predictions. This approach allowed the model to account for future uncertainties in temperature, leading to more accurate peak hour forecasts. The model was evaluated using Mean Squared Error (MSE) with a ± 1 hour adjustment to accommodate slight timing deviations in predictions. The results were compared with naive forecasting models, which confirmed the superior performance of the SVR model in most cases.

Regions such as Southeastern Massachusetts and Connecticut exhibited the lowest errors, demonstrating the model's effectiveness in these areas. However, some regions, like Maine, posed greater challenges due to unique weather and demand patterns, suggesting that further refinements could be explored to better capture the complexities of these areas.

In addition to point forecasts, the thesis also evaluated the performance of probabilistic forecasts using the Brier Score. The results showed that Southeastern Massachusetts had the best performance, while Northeast Massachusetts and Maine experienced the highest errors, highlighting areas where further improvements could be made.

A potential future direction for this research could involve developing a meteorological station selection procedure. This would entail creating model combinations for each load zone with various meteorological stations to determine which station provides the best predictive performance for each zone. By systematically evaluating the accuracy of these model combinations, it would be possible to optimize the choice of meteorological data used for forecasting in different regions. This approach could significantly enhance model performance by ensuring that the most relevant weather data is utilized for each specific load zone.

Additionally, future research could explore incorporating other features such as humidity and economic indicators to further improve prediction accuracy. Integrating renewable energy data and investigating the effects of extreme weather events on electricity demand could provide valuable insights for grid management. Expanding the use of more advanced machine learning techniques, such as deep learning models, may also yield improvements in peak hour forecasting.

The code and datasets utilized for this thesis can be found at https://github.com/SofiaGianelli/Final-Master-Thesis

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